合成高分子课堂笔记（二）

Commercialized and Engineering Polymers

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目录

1 基本概念与分类（Basic Concepts & Classification）

1.1 关键定义（Key Definitions）

- **Monomer / 单体**: A small molecule that can react chemically with other monomers to form a polymer chain. （能与其他单体反应形成聚合物链的小分子。）

- **Polymer / 聚合物**: A large macromolecule composed of many repeating structural units (monomers) connected by covalent bonds. （由许多重复结构单元通过共价键连接而成的大分子。）

- **Degree of Polymerization (DP) / 聚合度**: The number of monomer units in a polymer chain. $DP = M_n/M_0$ where M_n is number-average molecular weight, M_0 is monomer molecular weight. （聚合物链中单体单元的数目。）

- **Number-average molecular weight (M_n) / 数均分子量**: Total weight of all polymer chains divided by the number of chains. （按聚合物链的数量平均的分子量。）

- **Weight-average molecular weight (M_w) / 重均分子量**: Weighted by the mass of chains; always larger than M_n for polydisperse samples. （按链的质量加权平均的分子量，对于多分散样品通常大于 M_n。）

- **Polydispersity index (PDI)**: M_w/M_n; indicates the breadth of molecular weight distribution. （分子量分布宽度指数。）

- **End group / 端基**: The functional group at the end of a polymer chain, important for characterization and further reactions. （聚合物链末端的官能团。）

- **Crosslinking / 交联**: Covalent bonds that connect adjacent polymer chains, forming a three-dimensional network. High crosslink density gives thermosets; low density gives elastomers. （相邻聚合物链之间的共价连接，形成三维网络。高交联密度 → 热固性塑料；低交联密度 → 弹性体。）

- **Thermoplastic / 热塑性塑料**: Polymers that soften when heated and harden when cooled; process is reversible; recyclable. Chains are linear or branched, not crosslinked.（加热软化、冷却硬化，过程可逆，可回收。链为线形或支化，无交联。）
- **Thermoset / 热固性塑料**: Polymers that undergo irreversible crosslinking during curing; once set, they cannot be remelted or reshaped. （固化过程中发生不可逆交联，一旦成型不能再熔化或重塑。）
- **Elastomer / 弹性体**: Polymers with low crosslinking density that can be stretched to high strains and return to original shape; rubber-like. （低交联密度聚合物，可高倍拉伸并恢复原状，类似橡胶。）

1.2 聚合物拓扑结构（Polymer Topology）

- **Linear polymer / 线形聚合物**: Chains without branches; chains slide easily → thermoplastic. Example: HDPE. （无支链，链易滑动 → 热塑性。）
- **Branched polymer / 支化聚合物**: Linear backbone with side chains; more heat needed to soften; still thermoplastic. Example: LDPE. （主链带有侧链，需要更多热量软化，仍为热塑性。）
- **Crosslinked network / 交联网络**: Adjacent chains covalently bonded → thermoset; cannot be reformed. Example: epoxy, vulcanized rubber. （相邻链共价键合 → 热固性，不能重塑。例：环氧树脂，硫化橡胶。）
- **Block copolymer / 嵌段共聚物**: A polymer consisting of long sequences (blocks) of one monomer followed by blocks of another monomer. Example: ABA triblock copolymer. （由一种单体的长链段与另一种单体的长链段交替组成。例：ABA。）
- **Graft copolymer / 接枝共聚物**: A backbone of one type of repeat unit with branches (grafts) consisting of another repeat unit. （一种重复单元构成主链，另一种重复单元构成侧链“接枝”。）
- **Random copolymer / 无规共聚物**: Two monomers randomly distributed along the chain. （两种单体沿链随机分布。）
- **Alternating copolymer / 交替共聚物**: Monomers alternate perfectly: ABABAB...（单体完美交替：ABABAB⋯）

1.3 热性能分类（Thermal Properties）

Thermoplastic, softens when heating, can be reshaped, recyclable. Thermoset, remains rigid, cannot reshape, non-recyclable.

1.4 天然与合成聚合物（Natural vs Synthetic Polymers）

- **Natural polymers / 天然聚合物**: Cotton (纤维素), starch (淀粉), natural rubber (聚异戊二烯), silk (丝蛋白), cellulose (纤维素), wool (羊毛), DNA, proteins (蛋白质).
- **Synthetic polymers / 合成聚合物**: Plastic bags, polyester fiber, synthetic rubber (SBR, etc.), polyurethane foams, polyethylene, nylon, PVC, PS, PET, ABS, PC.
- **Raw materials / 原料来源**: Crude oil or natural gas → petrochemicals (ethylene, propylene, styrene, etc.). （原油或天然气 → 石化产品（乙烯、丙烯、苯乙烯等）。）

1.5 聚合物应用分类（Application-based Classification）

- Plastic（塑料）, Fiber（纤维）, Rubber（橡胶）, Adhesive（粘合剂）, Surface coating（涂料）, Foam（泡沫）.

2 聚合物结构层次（Levels of Polymer Structure）

1. **Atomic level (原子级)**: Chemical structure of the repeat unit. Single bonds give flexible chains (e.g., PE); rigid aromatic rings give stiffness (e.g., Kevlar, PET). （重复单元的化学结构。单键 → 柔顺链；刚性芳环 → 刚性链。）
2. **Molecular level (分子级)**: Topology (linear, branched, network, helical), degree of polymerization, copolymer type. （拓扑结构、聚合度、共聚物类型。）
3. **Crystallinity (结晶度)**:
   - Amorphous (无定形): No long-range order; exhibits glass transition temperature T_g.
   - Crystalline (结晶): Ordered domains; exhibits melting temperature T_m.
   - Semicrystalline (半结晶): Combination of crystalline and amorphous regions. Most common.
4. **Phase behavior (相行为)**: How two or more polymers mix or phase separate. （多种聚合物混合时的相容性或相分离。）

3 大宗商品聚合物（Commodity Polymers）

3.1 聚烯烃（Polyolefins）

3.1.1 聚乙烯（Polyethylene, PE）

Monomer: ethylene ($CH_2 = CH_2$). Polymerization: addition (free radical, Ziegler-Natta, metallocene).

- **LDPE (Low-Density Polyethylene / 低密度聚乙烯)**: Highly branched; low crystallinity (∼40-50%); $T_g \approx -125\,°C$, $T_m \approx 105 - 115\,°C$. Soft, flexible. Uses: squeeze bottles, film, tubing.
- **HDPE (High-Density Polyethylene / 高密度聚乙烯)**: Linear; high crystallinity (∼70-80%); $T_g \approx -125\,°C$, $T_m \approx 125 - 135\,°C$. Stiff, strong. Uses: milk jugs, pipes, containers.
- **UHMWPE (Ultra-High Molecular Weight Polyethylene / 超高分子量聚乙烯)**: Very high molecular weight (数百万 g/mol); excellent abrasion resistance; $T_m \approx 130 - 140\,°C$. Uses: artificial joints, ballistic vests.

3.1.2 聚丙烯（Polypropylene, PP）

Monomer: propylene ($CH_3CH = CH_2$). $T_g \approx 0\,°C$ (isotactic), $T_m \approx 160 - 170\,°C$. Uses: carpet fibers, ropes, pipes, food containers, automotive parts. （等规 PP 的半结晶性，刚性优于 PE。）

3.1.3 聚苯乙烯（Polystyrene, PS）

Monomer: styrene ($C_6H_5CH = CH_2$). Atactic PS: amorphous, $T_g \approx 100\,°C$, brittle. Uses: packaging foam (expanded PS), lighting panels, appliance components. （无规 PS 是无定形、脆性的。）

3.1.4 聚氯乙烯（Polyvinyl Chloride, PVC）

Monomer: vinyl chloride ($CH_2 = CHCl$). $T_g \approx 80\,°C$ (depends on plasticizer). Rigid PVC (unplasticized) and flexible PVC (with plasticizers). Uses: pipes, electrical wire insulation, raincoats, toys.

3.2 树脂识别码（Resin Identification Code）

Code	Polymer	Recyclable / 可回收
1	PETE (PET)	Yes
2	HDPE	Yes
3	PVC	Limited
4	LDPE	Sometimes
5	PP	Yes
6	PS	Limited
7	Other (PC, PLA, etc.)	Varies

表 1: Plastic recycling codes（塑料回收代码）

Note: Recyclable plastic automatically reused; proper sorting required. （可回收塑料不等于自动被再利用，需要正确分类。）

4 工程聚合物（Engineering Polymers）

4.1 聚对苯二甲酸乙二醇酯（Polyethylene Terephthalate, PET）

- Family: Polyester (ester linkage −COO− in main chain). （主链含酯键。）
- Monomers: Ethylene glycol (EG) + Terephthalic acid (TPA) or Dimethyl terephthalate (DMT). （乙二醇 + 对苯二甲酸或对苯二甲酸二甲酯。）
- Polymerization: Condensation (step-growth). （缩聚反应。）
- Thermal transitions: $T_g \approx 70-80\,°C$, $T_m \approx 250-260\,°C$.（PPT 提到软化温度 70°C，即接近 T_g。）
- Properties: High stiffness and strength (due to aromatic ring); semicrystalline when drawn; transparent when rapidly cooled; good gas barrier. （高刚性和强度（芳环贡献）；拉伸后半结晶；快速冷却时透明；优良的气体阻隔性。）
- Applications: Carbonated beverage bottles, textile fibers (polyester), food jars, Mylar films, face shields. （碳酸饮料瓶、涤纶纤维、食品罐、麦拉膜、防护面罩。）
- Limitation: Low T_g –not for hot-filling (>70°C). （玻璃化转变温度低，不适用于热灌装。）

4.2 聚酰胺（Polyamide, PA / Nylon）

- Family: Contains amide group (−CONH−). （含酰胺键。）
- Natural: Proteins (wool, silk). Synthetic: Nylon.
- **Nylon 6,6**:
  - Monomers: Hexamethylenediamine (C6 diamine) + Adipic acid (C6 diacid).（己二胺 + 己二酸。）
  - Polymerization: Condensation (eliminates water). （缩聚，失水。）
  - Thermal transitions: $T_g \approx 50 - 60\,°\mathrm{C}$, $T_m \approx 260 - 265\,°\mathrm{C}$.
  - Properties: High strength, toughness, abrasion resistance, good solvent resistance due to hydrogen bonding between chains. （高强度、韧性、耐磨性、良好的耐溶剂性（链间氢键）。）
  - History: DuPont (1935); first commercial use in toothbrushes (1938); women's stockings (1940).
- **Nylon 6** (from caprolactam via ring-opening polymerization): $T_g \approx 50 - 55\,°\mathrm{C}$, $T_m \approx 220 - 225\,°\mathrm{C}$.

4.3 聚碳酸酯（Polycarbonate, PC）

- Monomers: Bisphenol A + Phosgene (or diphenyl carbonate). （双酚 A+ 光气。）
- Polymerization: Condensation (step-growth).
- Thermal transitions: $T_g \approx 145 - 150\,°\mathrm{C}$; typically amorphous (no distinct T_m). （无定形，仅有玻璃化转变。）
- Properties: Transparent, very high impact resistance (shatterproof), high refractive index (1.586) → thinner lenses. （透明、极高抗冲击性（防碎）、高折射率 → 镜片更薄。）
- Applications: Eyeglass lenses, bulletproof glass, lightweight windows, electronic components. （眼镜镜片、防弹玻璃、轻质窗户、电子元件。）

4.4 丙烯腈-丁二烯-苯乙烯共聚物（Acrylonitrile Butadiene Styrene, ABS）

- Composition: 15-35% acrylonitrile, 5-30% butadiene, 40-60% styrene.

- Polymerization: Emulsion or bulk polymerization.
- Thermal transitions: T_g of styrene-acrylonitrile (SAN) matrix $\approx$ 105 °C; butadiene rubber phase provides low temperature toughness (butadiene $T_g \approx -85$ °C). Overall ABS is amorphous with $T_g \approx 105$ °C (no sharp T_m).
- Role of each component:
  - Acrylonitrile (丙烯腈): Chemical resistance, heat stability.
  - Butadiene (丁二烯): Toughness, impact strength (rubber toughening).
  - Styrene (苯乙烯): Rigidity, processability (easy injection molding).
- Properties: Opaque, impact-resistant, amorphous thermoplastic; good for 3D printing (FDM filament). （不透明、抗冲击、无定形热塑性塑料；适合 3D 打印。）
- Applications: LEGO bricks, automotive interior parts, home appliances, electronics housings. （乐高积木、汽车内饰、家电、电子设备外壳。）

4.5 聚氨酯（Polyurethane, PU）

- Monomers: Diisocyanate (e.g., MDI, TDI) + Polyol (diol or triol).
- Polymerization: Step-growth forming urethane linkage ($-\mathrm{NH}-\mathrm{CO}-\mathrm{O}-$).
- Can be thermoplastic (TPU) or thermoset (crosslinked foams).
- T_g varies widely from -50 °C to $+150$ °C depending on soft/hard segment ratio.
- Applications: Flexible foams (mattresses), rigid foams (insulation), elastomers (wheels, seals), coatings, adhesives, Spandex fibers.

4.6 纤维素及其衍生物（Cellulose and Derivatives）

- Natural polymer from wood, cotton; polysaccharide of glucose; extensive hydrogen bonding → high crystallinity. （天然聚合物，来自木材、棉花；葡萄糖多糖；广泛的氢键 → 高结晶度。）
- Derivatives: Cellulose nitrate (early plastic), cellulose acetate (textiles), hydroxyethylcellulose (HEC).
- Hydroxyethylcellulose: Water-soluble, non-crystalline; used as thickener in shampoos to suspend dirt. （水溶性、非结晶；用作洗发水增稠剂，悬浮污垢。）

5 高性能聚合物（High Performance Polymers）

表 2: High Performance Polymers / 高性能聚合物

Polymer	$\mathbf{T_g}$ (°C)	$\mathbf{T_m}$ (°C)	**Key Properties & Applications**
PEEK (聚醚醚酮)	∼143	∼343	High temp resistance (>250 °C continuous), chemical resistance; aerospace, medical implants.
PTFE (聚四氟乙烯)	∼115	∼327	Very low friction, non-stick (Teflon); coatings, gaskets, bearings.
PDMS (聚二甲基硅氧烷)	∼-123	– (amorphous)	Silicone elastomer, biocompatible; soft lithography, contact lenses.
Kevlar (聚对苯二甲酰对苯二胺)	∼360 (decomposes)	No T_m (decomposes)	High tensile strength, aromatic rigid chains; bulletproof vests, composites.
PI (聚酰亚胺)	>250	– (amorphous or semicrystalline)	High thermal stability (up to 400°C); Kapton tape, flexible circuits.
PVDF (聚偏氟乙烯)	∼-40	∼170	Piezoelectric, chemical resistant; sensors, membranes.

6 催化剂与聚合方法（Catalysts & Polymerization Methods）

- **Ziegler-Natta catalysts** (齐格勒-纳塔催化剂): Transition metal (Ti, Zr) + organoaluminum compound. Enable stereoregular polymerization (isotactic PP) and high molecular weight at mild conditions. Important for HDPE, isotactic PP.

- **Metallocene catalysts** (茂金属催化剂): Single-site catalysts giving very narrow molecular weight distribution.
- **Free radical polymerization** (自由基聚合): Used for LDPE (high pressure), PS, PVC.

7 塑料回收与污染（Plastic Recycling & Pollution）

- Reduce single-use plastics.
- Recycle according to resin codes (1, 2, 5 are most recycled).
- For unrecyclable plastics: chemical breakdown (pyrolysis, hydrolysis, biodegradation).（对于不可回收塑料，化学分解——热解、水解、生物降解。）
- Bioplastics (生物塑料): From renewable resources (corn starch, cellulose). Not all bioplastics are biodegradable; some are durable (e.g., bio-PET).（来自可再生资源，并非所有生物塑料都可生物降解，有些是耐用的。）

附录 1：常用聚合物中英文对照及热性能速查表

Polymer		**T_g** (°C)	**T_m** (°C)	**Application Example**
LDPE	低密度聚乙烯	-125	105-115	Squeeze bottles
HDPE	高密度聚乙烯	-125	125-135	Milk jugs, pipes
UHMWPE	超高分子量聚乙烯	-125	130-140	Artificial joints
PP (isotactic)	等规聚丙烯	0	160-170	Carpet fibers, cups
PS (atactic)	无规聚苯乙烯	100	– (amorphous)	Foam packaging
PVC (rigid)	硬质聚氯乙烯	80	– (amorphous)	Pipes, window frames
PET	聚对苯二甲酸乙二醇酯	70-80	250-260	Beverage bottles, fibers
Nylon 6,6	尼龙 66	50-60	260-265	Textiles, brushes, gears
Nylon 6	尼龙 6	50-55	220-225	Carpets, fibers

PC	聚碳酸酯	145-150	amorphous	Eyeglass lenses, bulletproof glass
ABS	ABS 树脂	∼105	amorphous	LEGO bricks, electronics
PEEK	聚醚醚酮	143	343	Aerospace, medical implants
PTFE	聚四氟乙烯	115	327	Non-stick coatings
Kevlar	凯夫拉（芳纶）	360 (decomp.)	no T_m	Bulletproof vests
PDMS	聚二甲基硅氧烷	-123	amorphous	Soft contact lenses, sealants
PI	聚酰亚胺	>250	varies	Kapton tape, flexible PCBs

表 3: Comprehensive Polymer Datasheet

附录 2：中英文术语对照表

English	**中文**
Addition polymerization	加聚反应
Condensation polymerization	缩聚反应
Degree of polymerization (DP)	聚合度
Number-average molecular weight (M_n)	数均分子量
Weight-average molecular weight (M_w)	重均分子量
Polydispersity index (PDI)	多分散指数
End group	端基
Crosslinking	交联
Thermoplastic	热塑性塑料
Thermoset	热固性塑料
Elastomer	弹性体
Linear polymer	线形聚合物
Branched polymer	支化聚合物
Block copolymer	嵌段共聚物
Graft copolymer	接枝共聚物
Random copolymer	无规共聚物
Alternating copolymer	交替共聚物

Amorphous	无定形的
Crystalline	结晶的
Semicrystalline	半结晶的
Glass transition temperature (T_g)	玻璃化转变温度
Melting temperature (T_m)	熔融温度
Polyethylene (PE)	聚乙烯
Low-density PE (LDPE)	低密度聚乙烯
High-density PE (HDPE)	高密度聚乙烯
Ultra-high molecular weight PE (UHMWPE)	超高分子量聚乙烯
Polypropylene (PP)	聚丙烯
Polystyrene (PS)	聚苯乙烯
Polyvinyl chloride (PVC)	聚氯乙烯
Polyethylene terephthalate (PET)	聚对苯二甲酸乙二醇酯
Polyamide (PA) / Nylon	聚酰胺 / 尼龙
Nylon 6,6	尼龙 66
Polycarbonate (PC)	聚碳酸酯
Acrylonitrile butadiene styrene (ABS)	丙烯腈-丁二烯-苯乙烯共聚物
Polyurethane (PU)	聚氨酯
Polyether ether ketone (PEEK)	聚醚醚酮
Polytetrafluoroethylene (PTFE)	聚四氟乙烯
Polydimethylsiloxane (PDMS)	聚二甲基硅氧烷
Kevlar	凯夫拉（芳纶）
Polyimide (PI)	聚酰亚胺
Polyvinylidene fluoride (PVDF)	聚偏氟乙烯
Resin identification code	树脂识别码
Bioplastic	生物塑料
Cellulose	纤维素
Ziegler-Natta catalyst	齐格勒-纳塔催化剂
Metallocene catalyst	茂金属催化剂
Free radical polymerization	自由基聚合
Emulsion polymerization	乳液聚合
Interfacial polymerization	界面聚合

重点提示

1. **区分热塑性与热固性**: 本质是线形/支化（可逆）vs 交联网络（不可逆）。

2. **写出聚合物的重复单元结构**: 例如 PET 的重复单元为 $-\mathrm{CO}-\mathrm{C_6H_4}-\mathrm{COO}-\mathrm{CH_2CH_2O}-$。

3. **解释结构与性能关系**: 芳环 → 刚性/高 T_m；氢键（尼龙）→ 强度；交联 → 弹性或热固性。

4. **树脂代码 1-7**: 1(PET), 2(HDPE), 3(PVC), 4(LDPE), 5(PP), 6(PS), 7(其他)。

5. **热转变数据**: PET ($T_g \sim 70°\mathrm{C}$) 不适合热灌装；PC 的 T_g 高，可用作耐热透明材料；PP 的 T_m 高于 PE。

6. **尼龙合成**: 尼龙 66 为缩聚，尼龙 6 为开环聚合。

7. **ABS 三组分功能**: 丙烯腈 (耐化学品/热稳定性)、丁二烯 (韧性)、苯乙烯 (刚性/加工性)。